Linux File Permissions and Access Control

Course: Google Cybersecurity Professional Certificate
Project Type: Linux Security and Access Control
Environment: Linux Bash Shell (Simulated Lab)

1. Project Description

In this project, I used Linux commands to inspect and modify file and directory permissions. I practiced identifying file owners, groups, hidden files, and different levels of read, write, and execute access.

The goal was to understand how Linux permissions can be used to restrict unauthorized access.

2. Check File and Directory Details

I used `ls -l` to view file permissions, ownership, groups, file sizes, and modification dates.

I also used `ls -la` to display hidden files, including `.project_x.txt`.

```
researcher2@3594b90052f9:~$ ls
projects
researcher2@3594b90052f9:~$ ls -l
total 4
drwxr-xr-x 3 researcher2 research_team 4096 Sep  5 10:43 projects
researcher2@3594b90052f9:~$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Sep  5 12:15 .
drwxr-xr-x 1 root        root          4096 Sep  5 10:43 ..
-rw------- 1 researcher2 research_team   15 Sep  5 12:18 .bash_history
-rw-r--r-- 1 researcher2 research_team  220 Mar 27  2022 .bash_logout
-rw-r--r-- 1 researcher2 research_team 3574 Sep  5 10:43 .bashrc
-rw-r--r-- 1 researcher2 research_team 3574 Sep  5 10:43 .profile
drwxr-xr-x 3 researcher2 research_team 4096 Sep  5 10:43 projects
researcher2@3594b90052f9:~$ cd projects
researcher2@3594b90052f9:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Sep  5 10:43 .
drwxr-xr-x 3 researcher2 research_team 4096 Sep  5 12:15 ..
-rw--w---- 1 researcher2 research_team   46 Sep  5 10:43 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Sep  5 10:43 drafts
-rw-rw-rw- 1 researcher2 research_team   46 Sep  5 10:43 project_k.txt
-rw-r----- 1 researcher2 research_team   46 Sep  5 10:43 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Sep  5 10:43 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Sep  5 10:43 project_t.txt
researcher2@3594b90052f9:~/projects$
```

3. Understanding Permission Strings

Linux permissions are divided into three categories: user (owner), group, and others.

For example, `-rw-rw-r--` means:

- Owner: Read and write
- Group: Read and write
- Others: Read only

The first character indicates the file type, while the remaining nine characters represent read (`r`), write (`w`), and execute (`x`) permissions.

4. Changing File Permissions

I used the `chmod` command to modify permissions for specific files.

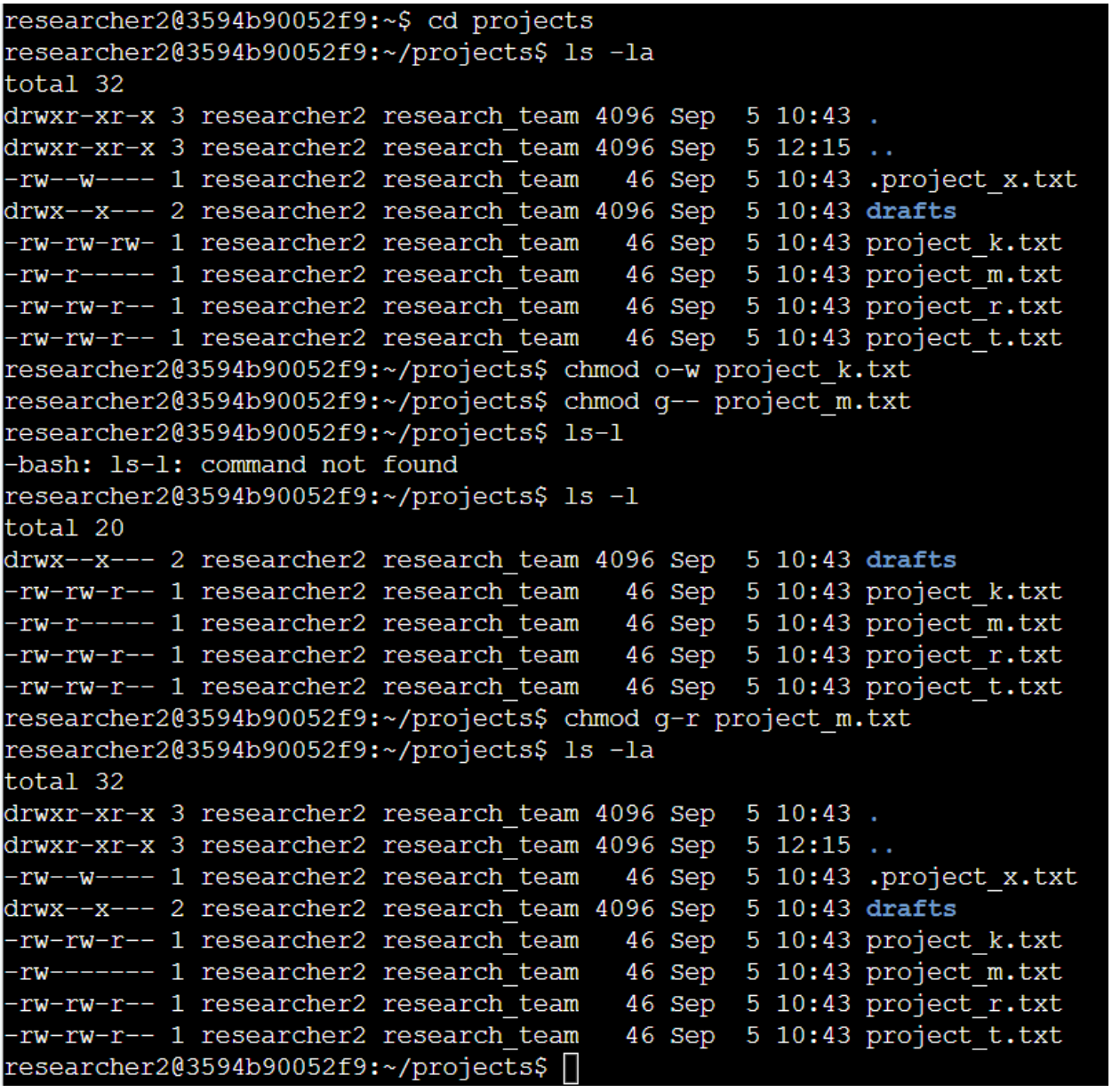
Commands used:

```
chmod o-w project_k.txt
```

```
chmod g-r project_m.txt
```

The first command removes write permission from others, while the second removes read permission from the group.

I used `ls -l` and `ls -la` to inspect the permissions after making changes.



5. Changing Permissions on a Hidden File

I identified the hidden file `.project_x.txt` using `ls -la`.

I initially encountered an error while attempting to combine permission changes in one command.

I corrected the syntax by executing separate commands:

```
chmod g-w .project_x.txt

chmod u-w .project_x.txt

chmod g+r .project_x.txt
```

These commands removed write permissions from the owner and group while allowing the group to read the file.

I then used `ls -la` to verify the updated permissions.

```
researcher2@3594b90052f9:~/projects$ chmod u-w, g-w
chmod: invalid mode: 'u-w,'
Try 'chmod --help' for more information.
researcher2@3594b90052f9:~/projects$ chmod u-w, g-w .project_x.txt
chmod: invalid mode: 'u-w,'
Try 'chmod --help' for more information.
researcher2@3594b90052f9:~/projects$ chmod g-w .project_x.txt
researcher2@3594b90052f9:~/projects$ chmod u-w .project_x.txt
researcher2@3594b90052f9:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Sep  5 10:43 .
drwxr-xr-x 3 researcher2 research_team 4096 Sep  5 12:15 ..
-r-------- 1 researcher2 research_team   46 Sep  5 10:43 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Sep  5 10:43 drafts
-rw-rw-r-- 1 researcher2 research_team   46 Sep  5 10:43 project_k.txt
-rw------- 1 researcher2 research_team   46 Sep  5 10:43 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Sep  5 10:43 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Sep  5 10:43 project_t.txt
researcher2@3594b90052f9:~/projects$ chmod g+r
chmod: missing operand after 'g+r'
Try 'chmod --help' for more information.
researcher2@3594b90052f9:~/projects$ chmod g+r .project_x.txt
researcher2@3594b90052f9:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Sep  5 10:43 .
drwxr-xr-x 3 researcher2 research_team 4096 Sep  5 12:15 ..
-r--r----- 1 researcher2 research_team   46 Sep  5 10:43 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Sep  5 10:43 drafts
-rw-rw-r-- 1 researcher2 research_team   46 Sep  5 10:43 project_k.txt
-rw------- 1 researcher2 research_team   46 Sep  5 10:43 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Sep  5 10:43 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Sep  5 10:43 project_t.txt
researcher2@3594b90052f9:~/projects$
```

6. Directory Permissions

I examined the permissions of the `drafts` directory.

The directory initially had the permission string `drwx--x---`, which allowed the owner full access and the group execute-only access.

This exercise helped me understand how directory permissions control access to files and directories.

My available terminal screenshots show the directory permissions being inspected but do not show a completed permission change for `drafts`.

7. Troubleshooting and Lessons Learned

During the exercise, I encountered errors involving incorrect command syntax, including an invalid `chmod` mode and a mistyped `ls` command.

I corrected the commands and checked their output to understand whether the intended permission changes were applied.

This helped me become more comfortable using the Linux terminal and understanding how small syntax errors can affect commands.

8. Summary

This project gave me hands-on experience working with Linux file and directory permissions.

I learned how to interpret permission strings, identify owners and groups, work with hidden files, and use `chmod` to modify access permissions.

I also gained experience troubleshooting command errors and verifying results through terminal output.

This project was completed as part of the Google Cybersecurity Professional Certificate in a simulated Linux environment. The screenshots show my original terminal work, including errors and corrections.